

in the flow. There is also a transitional stage between laminar and turbulent flows, in which the dye stream will wander about and show intermittent bursts of mixing, followed by a more laminar behavior.

The Reynolds number (Re), provides a useful way of characterizing the flow. It is defined as:

$$Re = \frac{vd}{\nu} \quad (1)$$

where (ν) is the kinematic viscosity of the water (Figure 7.2), v is the mean flow velocity and d is the diameter of the pipe.

The Reynolds number is a dimensionless parameter that is the ratio of the inertial (destabilizing) force to the viscosity (stabilizing) force. As Re increases, the inertial force becomes relatively larger, and the flow destabilizes and becomes fully turbulent.

The Reynolds experiment determines the critical Reynolds number for pipe flow at which laminar flow ($Re < 2000$) becomes transitional ($2000 < Re < 4000$) and the transitional flow becomes turbulent ($Re > 4000$). The advantage of using a critical Reynolds number, instead of critical velocity, is that the results of the experiments are applicable to all Newtonian fluid flows in pipes with a circular cross-section.

Temperature (degree C)	Kinematic viscosity ν (m ² /s)	Temperature (degree C)	Kinematic viscosity ν (m ² /s)
0	1.793E-06	25	8.930E-07
1	1.732E-06	26	8.760E-07
2	1.674E-06	27	8.540E-07
3	1.619E-06	28	8.360E-07
4	1.522E-06	29	8.180E-07
5	1.520E-06	30	8.020E-07
6	1.474E-06	31	7.850E-07
7	1.429E-06	32	7.690E-07
8	1.386E-06	33	7.530E-07
9	1.346E-06	34	7.380E-07
10	1.307E-06	35	7.240E-07
11	1.270E-06	36	7.110E-07
12	1.235E-06	37	6.970E-07
13	1.201E-06	38	6.840E-07
14	1.169E-06	39	6.710E-07
15	1.138E-06	40	6.580E-07
16	1.108E-06	45	6.020E-07
17	1.080E-06	50	5.540E-07
18	1.053E-06	55	5.110E-07
19	1.027E-06	60	4.760E-07
20	1.002E-06	65	4.430E-07
21	9.780E-07	70	4.130E-07
22	9.550E-07	75	3.860E-07
23	9.330E-07	80	3.630E-07
24	9.110E-07	85	3.420E-07

Figure 7.2: Kinematic Viscosity of Water at Atmospheric Pressure.